

CURBCOUNT · DESIGN DECISIONS REGISTER

CurbCount design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the build plan; open decisions gathered from the review note and the decision records

CurbCount design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note; this register is the index Amish works from. The build plan (`docs/05-build-plan.md` , CBC-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Decisions still to be made.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction (saddles, bands through slots, FieldNode constructable core, bolted arm, head fixing and window frame, bolted pole-top mount, cable routes, notice ties)	Accept; or change individual items	Accept	The whole build plan	CBC-DDR-003, A2
2	First co-design partner and street	City transport department, residents' group or university	None made	Pole size and band size; climate (a hot city brings the radar variant forward)	CBC-DDR-002, O1
3	Sensor port pinout, port A's supply voltage and the calibration jumper, agreed with FieldNode	I2C and a switched 3.3 V rail on port A's five pins (the build plan fits a 3.3 V load switch in place of FieldNode's port A boost converter); a jumper-sense pin on FieldNode's service header	Agree with the FieldNode project	Which cores of the sensor cable go to which breakout pin; the port A supply module (build plan section 3.3)	REVIEW 2026-09-25 cross-repo actions; FND-DDR-001, O2
4	Pole top drawn 560 mm lower in the photoreal renders	Accept as render-only layout; or render at the installed height	Accept as render-only	None (renders only)	REVIEW 2026-09-26, item 1
5	Notice plate drawn at 4.15 m in the renders (installed at 2.6 m)	Accept as render-only; or render at 2.6 m	Accept as render-only	None (renders only)	REVIEW 2026-09-26, item 2
6	Notice plate wording	Adopt the proposed wording ("PRIVACY-SAFE COUNTER", what is counted, no images taken or stored, only counts sent, repository link); or reword	Adopt. The fixing is now set by the design for construction (two ties through slots)	The printed face of the notice plate	REVIEW 2026-09-26, item 3
7	Status light on the sensor head	Drop it; or keep it as a public "counting" cue at about 1 mW	Drop it	None in the build plan (no light is fitted)	REVIEW 2026-09-26, item 4
8	"COUNTS ONLY" plaque on the head and the enclosure label	Print or emboss the plaque in the housing and treat the label as part of BOM line 1; or separate parts	Print and treat as BOM line 1	Head housing print (an optional embossed plaque)	REVIEW 2026-09-26, item 5
9		Superseded by the window frame and M5		None beyond item 1	REVIEW 2026-09-26,

#	Decision needed	Options	Recommendation	Affects in the build	Source
	Head window bezel and head bolts shown in the appearance model	bolts of the design for construction; update the appearance model to match	Close as superseded once item 1 is accepted		item 6; CBC-DDR-003, P4

■ To confirm when parts are bought

Table 2. Things to check against the real part before it is used.

#	What to confirm	Why it matters	Source
1	The solar panel's frame has a flat back lip at least 12 mm wide at its short ends	The rail plate bolts through it; without it the panel cannot be fixed as drawn	CBC-DDR-003, P5
2	The enclosure model, its boss spacing (110 x 160 mm assumed) and its lug kit	They set the internal plate holes and the lug screw holes in the saddle plate	FND-DDR-003; CBC-DDR-003, P2
3	Band clamps that close on the loop for the site pole: about 470 mm on a 114 mm pole (330 mm on 60 mm, 545 mm on 140 mm)	A worm-drive band has a limited range; the wrong size will not tighten	CBC-DDR-003, P1
4	The sleeve tube's bore is at least 122 mm, and the cap disc fits it closely	The sleeve must clear a 114.3 mm pole by 3 to 4 mm all round	CBC-DDR-003, P5
5	The band torque that gives 1,000 N of preload	The twist calculation assumes that preload	CBC-CAL-001 [H3]
6	The sensor cable has five cores of about 0.25 mm ² , a moulded M12 A-coded plug and an open end	The open end goes through the head gland to the breakout	CBC-DDR-003, P6
7	Prices of every BOM line	The estimate is \$1.00 under the \$275 value-engineering target	CBC-CAL-001 [J1]

■ Value engineering

Value-engineering target: USD 275 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 274.00 (USD 1.00 under the target). The estimate rests on indicative prices.

Main cost drivers: the FieldNode share (lines 1 to 3, 14 and part of line 13) at USD 105.00, FieldNode's own panel (line 4) at USD 14, and the parts added for construction together with FieldNode's repricing of its core, which took the total from USD 253.00 to USD 274.00. The ESP32-S3 fallback head would cost about USD 311.00, over the target.

Savings worth trying: leaving port B out of CurbCount's core (about USD 6 less, departing from the standard FieldNode core), and confirming prices when parts are bought.

■ Decisions made

Table 3. Decisions made, with the words recorded.

Date	Decision	Decided by	Record
2026-09-25	Batch taken through TRL 3, nothing past it	Amish: "you know the drill, nothing gets past TRL 3"	REVIEW 2026-09-25 (TRL 3)
2026-09-25	TRL 2 items D1 to D7: \$275 value-engineering target, thermal array first with radar as the hot-climate variant, FieldNode high-load variant raised with FieldNode, processor study, 15-minute bins and three classes, jumper-only calibration, public notice plate	Amish: "i accept all your recommendations, go with them across all repos."	CBC-DDR-001, CBC-DDR-002
2026-09-25	Tracking on FieldNode's STM32WL as the baseline with the ESP32-S3 head as fallback; one cell and the 6 W panel; R4 restated in px/m; R7 for temperate sites; R11 kept at 6 kg; 10-byte record	Amish, same instruction	CBC-DDR-002, items 3 to 8
2026-09-30	Build plan format approved for all repos; outstanding decisions kept out of the build plan, in this register	Amish: "this is the correct build plan ... this is a good quality document format. Extend this across all the other repos"	This register; CBC-BLD-001
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	CBC-DDR-003 (changes made under this instruction, open for review: open decision 1)